Orographic interpolation

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1 3D variables

1.1 Vertical discretization

The vertical levels is built with:

- Half-levels top-down from 0 to n_z
- Full-levels top-down from 0 to $n_z - 1$

The surface pressure P_s is defined at the last half-level (n_z). Hybrid coefficients a_k and b_k are defined at half-levels, to compute the pressure at half-levels:

$$P^h_k = a_k + b_k P_s \tag{1}$$

A full-levels, the temperature T^f_k is known, and the pressure P^f_k is given by:

$$P^f_k = \frac{P^h_k + P^h_{k+1}}{2} \tag{2}$$

1.2 Height calculation

The hydrostatic equation is:

$$\frac{\partial P}{\partial z} = -\rho g \tag{3}$$

where:

- ρ is the density
- g is the gravitational acceleration

and the ideal gas equation is:

$$P = \rho R^\star T \tag{4}$$

where $R^{\star}$ is the specific gas constant for air. Combining them, we get:

$$\frac{\partial P}{\partial z} = -\frac{Pg}{R^{\star}T} \tag{5}$$

which can be discretized into:

$$\delta z = -\frac{R^{\star}T}{Pg}\delta P \tag{6}$$

Integrating this equation upward from the surface, we get iteratively the altitude for each half-level, starting with $z^h_{n_z} = z_s$:

$$z^h_{k-1} = z^h_k - \frac{R^{\star}T^f_{k-1}}{P^f_{k-1}g}\left(P^h_{k-1} - P^h_k\right) \tag{7}$$

The height at full-levels is approximated by:

$$z^f_k = \frac{z^h_{k+1} + z^h_k}{2} \tag{8}$$

which is slightly inconsistent with the pressure at full levels.

1.3 Constant height coordinate

The constant height coordinate is defined by:

- the constant height surface: $\overline{z}_s = \min(z_s)$
- the constant height full-levels:

$$\overline{z}^f_k = \alpha_k \min(z^f_k) + (1-\alpha_k)\max(z^f_k) \tag{9}$$

  where α_k increases linearly from $\alpha_0 = 0$ to $\alpha_{n_z-1} = 1$.

The n_s constant half-levels are defined as:

$$\overline{z}^h_{n_s} = \overline{z}_s \tag{10a}$$

$$\overline{z}^h_k = \frac{\overline{z}^f_k + \overline{z}^f_{k-1}}{2} \text{ for } 0 < k < n_s \tag{10b}$$

$$\overline{z}^h_0 = 2\overline{z}^f_0 - \overline{z}^h_1 \tag{10c}$$

2 Surface pressure

2.1 Balanced increment

Equation (5) can be expressed as:

$$\frac{\partial \log P}{\partial z} = -\frac{g}{R^{\star}T} \tag{11}$$

which can be integrated from the constant height surface:

$$\log P(z) = \log P(\overline{z}_s) - \int_{\overline{z}_s}^{z} \frac{g}{R^\star T(z')} dz' \tag{12}$$

If the values of $\log P(\overline{z}_s)$ and $T(z')$ are perturbed with small perturbations $\delta \log P(\overline{z}_s)$ and $\delta T(z')$, then the resulting perturbation on $\log P(z)$ is:

$$\begin{aligned}\delta \log P(z) &= \Big(\log P(\overline{z}_s) + \delta \log P(\overline{z}_s)\Big) - \int_{\overline{z}_s}^{z} \frac{g}{R^\star \left(T(z') + \delta T(z')\right)} dz' - \log P(z) \\ &= \delta \log P(\overline{z}_s) - \int_{\overline{z}_s}^{z} \frac{g}{R^\star} \left(\frac{1}{(T(z') + \delta T(z'))} - \frac{1}{T(z')} \right) dz' \end{aligned} \tag{13}$$

Assuming that the temperature perturbation is small enough $\delta T(z') \ll T(z')$, then a first-order approximation yields:

$$\begin{aligned}\frac{1}{(T(z') + \delta T(z'))} - \frac{1}{T(z')} &= \frac{1}{T(z')} \left(\frac{1}{1 + \frac{\delta T(z')}{T(z')}} - 1 \right) \\ &\simeq -\frac{1}{T(z')} \frac{\delta T(z')}{T(z')} \\ &\simeq -\frac{\delta T(z')}{T^2(z')} \end{aligned} \tag{14}$$

Thus:

$$\delta \log P(z) \simeq \delta \log P(\overline{z}_s) + \int_{\overline{z}_s}^{z} \frac{g}{R^\star T^2(z')} \delta T(z') dz' \tag{15}$$

The integration be can discretized on constant half-levels:

$$\delta \log \overline{P}^h_k \simeq \delta \log \overline{P}_s + \sum_{k'=k}^{n_s-1} \frac{g}{R^\star \overline{T}^{f2}_{k'}} \left(\overline{z}^h_{k'} - \overline{z}^h_{k'+1} \right) \delta \overline{T}^f_{k'} \tag{16}$$

2.2 Interpolation extrapolation

Assuming a constant lapse rate, the temperature relationship is given by:

$$T_2 = T_1 - L\left(z_2 - z_1\right) \tag{17}$$

where:

- T_1 and T_2 are the temperatures at altitude z_1 and z_2
- L is the constant lapse rate

Thus, the temperature at constant height full-levels below orography can be computed from the lowest model level:

$$\overline{T}^f_k = T^f_{n_z-1} - L\left(\overline{z}^f_k - z^f_{n_z-1}\right) \tag{18}$$